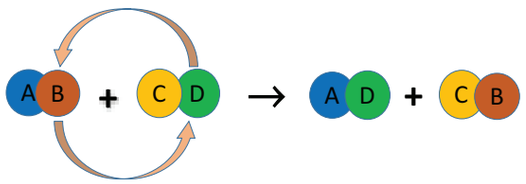
Activity 10.1

- Take about 50 ml of toilet cleaning acid in a beaker
- Place a small iron nail in it
- Wait for about 10 minutes
- Observe what happen in the beaker
- Do you recognize any change?
- Report your observations with the chemical equation.

(d) Double Displacement Reactions

When two compounds react, if their ions are interchanged, then the reaction is called double displacement reaction. The ion of one compound is replaced by the ion of the another compound. Ions of identical charges are only interchanged, i.e., a cation can be replaced by other cations. This reaction is also called '**Metathesis Reaction**'. The schematic representation of a double displacement reaction is given below:



In a double displacement reaction either one of the products must be either a precipitate or water. By this way, there are major classes of double displacement reactions. They are:

(i) Precipitation Reactions

(ii) Neutralization Reactions

(i) Precipitation Reactions

When aqueous solutions of two compounds are mixed, if they react to form an insoluble compound and a soluble compound, then it is called precipitation reaction. Because the insoluble compound, formed as one of the products, is a precipitate and hence the reaction is so called.

Table 10.1 Differences between combination and decomposition reactions

COMBINATION REACTIONS	DECOMPOSITION REACTIONS
One or more reactants combine to form a single product	A single reactant is decomposed to form one or more products
Energy is released	Energy is absorbed
Elements or compounds may be the reactants	Single compound is the reactant

When the clear aqueous solutions of potassium iodide and lead (II) nitrate are mixed, a double displacement reaction takes place between them.

$$Pb(NO_3)_{2(aq)} + 2KI_{(aq)} \rightarrow PbI_{2(s)}\downarrow + 2KNO_{3(aq)}$$

Potassium and lead displace or replace one other and form a yellow precipitate of lead (II) iodide as shown in Fig. 10.3.



Figure 10.3 Precipitation of PbI_2

Activity 10.2

- Take a pinch of silver nitrate crystals.
- Collect about 5 ml of tap water in a test tube.
- Add the silver nitrate crystals to water and shake well.
- Observe what happen in the test tube.
- Report your observations and what you infer from that?

(ii) Neutralization Reactions

In your lower classes, you have learned the reaction between an acid and a base. It is

10th_Science_Unit-10.indd 142 11-01-2020 10:37:00